

Homework 5

Pathri

Vidya

Praaveen - CS24BTECH

11047

Question 1:

(a) Quasiconvex function: Let $C \subseteq \mathbb{R}^n$ be a convex set. A function $f: C \rightarrow \mathbb{R}$ is called quasiconvex if $\forall x, y \in C$ and $\lambda \in [0, 1]$,

$$f(\lambda x + (1-\lambda)y) \leq \max\{f(x), f(y)\}$$

Equivalently, all sublevel sets $\{x \in C \mid f(x) \leq \alpha\}$ are convex $\forall \alpha \in \mathbb{R}$

(b)

$$f(x) = \begin{cases} 1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$

$$\text{Sublevel set } S_\alpha = \{x \in \mathbb{R} \mid f(x) \leq \alpha\}$$

① $\alpha < 0$: $S_\alpha = \emptyset$ (convex set)

② $0 \leq \alpha < 1$: $f(x) = 1 \forall x \neq 0$.

$$S_\alpha = \{0\} \text{ (convex set)}$$

③ $\alpha \geq 1$: $f(x) \leq 1 \leq \alpha \forall x \in \mathbb{R}$.

$$S_\alpha = \mathbb{R} \text{ — convex set}$$

Every sublevel set is convex. f is quasiconvex.

(c) $\Rightarrow$ Convexity test: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$
holds $\forall x, y, \lambda \in [0, 1]$.

$\Rightarrow$ Choose $x = -1, y = 0, \lambda = \frac{1}{2}$.

$$f\left(\frac{1}{2}(-1) + \frac{1}{2}(0)\right) = f(-0.5) = 1$$

$$\frac{1}{2}f(-1) + \frac{1}{2}f(0) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 = 0.5 \quad \left. \vphantom{\frac{1}{2}f(-1) + \frac{1}{2}f(0)} \right\} 1 > 0.5$$

$\Rightarrow$ Convex inequality test failed

$\Rightarrow f$ is not convex.

$\Rightarrow$ Quasi convexity does not imply convexity.

Quasi-convex functions can be discontinuous

Question 2:

(a) $\Rightarrow$ Convex conjugate of a function $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$

is defined as $f^*(y) = \sup_{x \in \text{dom } f} (y^T x - f(x))$,
 $y \in \mathbb{R}^n$

$\Rightarrow$ For 1-dimensional case, $n=1$,

$$f^*(y) = \sup_{x \in \mathbb{R}} (yx - f(x))$$

(b) $\Rightarrow f(x) = e^{-x}, x \in \mathbb{R}$

$$\Rightarrow f^*(y) = \sup_{x \in \mathbb{R}} (yx - e^{-x})$$

$$\Rightarrow g(x) = yx - e^{-x} \quad \therefore g'(x) = y + e^{-x}$$

$\Rightarrow$ ① $\forall y > 0, g'(x) > 0 \forall x, g$ is strictly increasing.

As $x \rightarrow \infty, g(x) \rightarrow \infty, f^*(y) = +\infty$

② $\forall y = 0, g(x) = -e^{-x} < 0 \forall x.$

$$\lim_{x \rightarrow +\infty} g(x) = 0, \quad f^*(y) = 0.$$

③ $\forall y < 0, g'(x) = 0 \rightarrow e^{-x} = -y, x = -\ln(-y).$

$g''(x) = -e^{-x} < 0$ so critical point gives maximum

$$f^*(y) = \sup_x y(-\ln(-y)) - e^{\ln(-y)} = y(1 - \ln(-y))$$

$$\rightarrow y(1 - \ln(-y)) \text{ if } y < 0$$

$$f^*(y) = \begin{cases} \rightarrow 0 & \text{if } y = 0 \end{cases}$$

$$\rightarrow +\infty \text{ if } y > 0$$

Question 3:

(a) $\Rightarrow$ For function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, sublevel set at level $\alpha \in \mathbb{R}$ is

$$S_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$$

(b) $\Rightarrow f(x_1, x_2) = \sqrt{x_1^2 + x_2^2} - x_1, (x_1, x_2) \in \mathbb{R}^2$

$\Rightarrow f$ is quasiconvex if all its sublevel sets are convex.

① $\alpha < 0$: $\sqrt{x_1^2 + x_2^2} \geq |x_1| \geq x_1, f(x) \geq 0.$

$f(x) \leq \alpha < 0$ — no solⁿ, $S_\alpha = \emptyset$ — convex set

② $\alpha = 0$: ~~$f(x) \leq 0$~~ $S_\alpha = \{x \mid f(x) \leq 0\}$

$f(x) \leq 0 \Rightarrow \sqrt{x_1^2 + x_2^2} \leq x_1.$

As we know, $\sqrt{x_1^2 + x_2^2} \geq |x_1|$, $x_1 \geq 0$ and $x_2 = 0$

$S_\alpha = \{(x_1, 0) \mid x_1 \geq 0\}$ — ray — convex set.

③ $\alpha > 0$: $\sqrt{x_1^2 + x_2^2} \leq x_1 + \alpha, x_1 + \alpha \geq 0.$

Squaring on both sides

$$x_1^2 + x_2^2 \leq x_1^2 + 2\alpha x_1 + \alpha^2$$

$$x_2^2 \leq 2\alpha x_1 + \alpha^2$$

$$x_1 \geq \frac{x_2^2}{2\alpha} - \frac{\alpha}{2}$$

$$\left. \begin{array}{l} x_1 \geq -\alpha \text{ because} \\ \frac{x_2^2}{2\alpha} - \frac{\alpha}{2} \geq -\alpha \\ \frac{x_2^2}{2\alpha} \geq -\frac{\alpha}{2} \\ \forall x_2 \end{array} \right\}$$

$$S_\alpha = \left\{ (x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq \frac{1}{2\alpha} x_2^2 - \frac{\alpha}{2} \right\}$$

S_α is epigraph of convex quadratic function

$$\phi(x_2) = \frac{1}{2\alpha} x_2^2 - \frac{\alpha}{2} \quad \text{so } S_\alpha \text{ is convex.}$$

⇒ All sublevel sets S_α are convex.

⇒ f is quasi convex.

Question 4:

(a) let $f: C \rightarrow [0, \infty)$ be a function defined on convex set $C \subseteq \mathbb{R}^n$.

⇒ f is log concave if $\log f$ is concave
i.e. $\forall x, y \in C$ and $t \in [0, 1]$

$$f(tx + (1-t)y) \geq f(x)^t f(y)^{1-t}$$

⇒ f is log convex if $\log f$ is convex i.e.

$\forall x, y \in C$ and $t \in [0, 1]$

$$f(tx + (1-t)y) \leq f(x)^t f(y)^{1-t}$$

⇒ When f takes value 0 or $\infty \rightarrow$ (extended reals)

(b) ⇒ $f(x) = \frac{1}{1+x^2} \quad x \in \mathbb{R}$

⇒ $g(x) = \log f(x) = -\log(1+x^2)$

⇒ $g'(x) = \frac{-2x}{1+x^2}$

⇒ $g''(x) = \frac{-2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2} = \frac{2(x^2-1)}{(1+x^2)^2}$

⇒ $g''(x) \geq 0 \iff |x| \geq 1$

⇒ $g''(x) \leq 0 \iff |x| \leq 1$

⇒ g'' changes sign on $\mathbb{R}$.

$\Rightarrow g$ is neither convex nor concave on whole real line

$\Rightarrow f$ is neither log-concave nor log-convex on $\mathbb{R}$.

Question 5:

(a) $\Rightarrow$ For a differentiable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the 1st order condition for quasi-convexity is:
 f is quasi-convex iff $\forall x, y \in \text{dom } f$

$$f(y) \leq f(x) \rightarrow \nabla f(x)^T (y - x) \leq 0$$

(b) $\Rightarrow f(x) = x^4, x \in \mathbb{R}$.

$$\Rightarrow f'(x) = 4x^3, f''(x) = 12x^2 \geq 0 \quad \forall x$$

f is convex.

$$\Rightarrow \forall y \in \mathbb{R}, \text{ conjugate } f^*(y) = \sup_{x \in \mathbb{R}} (yx - x^4)$$

Let $g(x) = yx - x^4$.

$$g'(x) = 0 \text{ for finding supremum}$$

$$y - 4x^3 = 0 \rightarrow x = \left(\frac{y}{4}\right)^{1/3}$$

~~$$f^*(y) = \sup_{x \in \mathbb{R}} yx - x^4 = y \left(\frac{y}{4}\right)^{1/3} - \left(\frac{y}{4}\right)^{4/3}$$~~

$$f^*(y) = y \left(\frac{y}{4}\right)^{1/3} - \left(\frac{y}{4}\right)^{4/3}$$

$$f^*(y) = 3 \cdot 4^{-4/3} |y|^{4/3}$$

$\Rightarrow$ Interpretation of $y = \nabla f(x)$:

For a convex differentiable function, gradient of the conjugate is inverse of gradient of f .

$$f'(x) = 4x^3. \quad y = 4x^3$$

$$\longrightarrow x = (f^*)'(y) = \left(\frac{y}{4}\right)^{1/3}$$

Conjugacy relation $f(x) + f^*(y) = xy$ holds.

Condition $y = \nabla f(x)$ is equivalent to

$$x = \nabla f^*(y)$$